

Vaccination Against Poly-N-Acetylglucosamine Decreases Neutrophil Activation and Gvhd While Maintaining Microbial Diversity

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Allogeneic hematopoietic cell transplantation (allo-HCT) is a curative therapy for different malignant and non-malignant hematologic diseases. Major life-threatening complications after allo-HCT are acute graft-versus-host disease (aGVHD) and severe infections. The incidence of GVHD after allo-HCT remains high, despite prophylactic immunosuppressive medication. According to the CIBMTR database, 60% of patients undergoing allo-HCT develop at least grade II-IV aGVHD. Microbial invasion into the intestinal mucosa after allo-HCT triggers the activation of neutrophil granulocytes (neutrophils) and requires antibiotic treatment to prevent sepsis. However, antibiotics lead to a loss of microbial diversity, which is connected to a higher incidence of aGVHD. Anti-microbial therapies which eliminate invading bacteria and diminish neutrophil-mediated damage without reducing the diversity of the microbiota are therefore highly desirable. A potential solution for this could be the use of anti-microbial antibodies that target and mark invading pathogens resulting in their elimination by innate immune cells.

Therefore we investigated the potency of both active and passive immunization against the conserved microbial surface polysaccharide Poly-*N*-acetylglucosamine (PNAG) which is expressed on various pathogens in a murine aGVHD model. We could detect reduced aGVHD related mortality in mice which underwent treatment with an antibody against PNAG (anti-PNAG) or were actively immunized against PNAG. Sequencing analysis of the microbiota before and after allo-HCT showed that anti-PNAG treatment did not have any effect luminal microbial diversity. Mechanistically, we could show that anti-PNAG antibody treatment caused more abundant neutrophil recruitment to the intestinal submucosa. This supports the concept that neutrophils are able to effectively eliminate any opsonized bacteria in this tissue, leading to reduced local inflammation. In agreement with this we observed lower intestinal inflammation and reduced myeloperoxidase activity as well as proliferation of neutrophils in the small intestine of mice treated with anti-PNAG antibody.

We demonstrate the potency of targeting PNAG as a novel antimicrobial treatment option in aGVHD by reducing uncontrolled neutrophil activation and thereby interfering with aGVHD without affecting commensal intestinal microbial diversity. Anti-PNAG-antibodies enhance anti-microbial immunity, while reducing the activation of neutrophil-mediated downstream immunopathology which is a novel approach to reduce the severity of GVHD. This could be translated into a clinical application given the modest toxicity profile of vaccination and the availability of fully human anti-PNAG antibodies, both of which have been tested in phase 1 trials in humans (ClinicalTrials.gov Identifier: NCT02853617).

Disclosures

Cywes-Bentley: *OneBiopharma, Inc.*: Honoraria. **Koenecke:** *Novartis*: Other: none. **Pier:**

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Author notes

*Asterisk with author names denotes non-ASH members.

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